Vinsea A V Singh

PhD Candidate · Cognitive and Computational Neuroscience

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Education

National Brain Research Centre (NBRC) Manesar *Haryana, India*
PhD Neuroscience *(Submitted)* *August 2020 - Present*
- Thesis: The Brain Before Perception: Trial-Wise Prestimulus Neural Dynamics Shape Multisensory Perception

National Brain Research Centre (NBRC) Manesar *Haryana, India*
MSc Neuroscience *August 2018 - July 2020*
- Dissertation: Characterizing variability of audiovisual speech perception based on prestimulus oscillatory features of electrophysiological brain signals [Link]

St. Xavier's College, Mumbai *Mumbai, India*
BSc Zoology and Biochemistry *June 2014 - March 2017*

Publications

Published

Singh, V. A., Roy, D., and Banerjee, A. (2025). Prestimulus EEG Studies in Multisensory Perception. *Electroencephalogram Signals and Cognition*, edited by Hema A. Murthy *(Book Chapter, in Press)*

Singh, V. A., Kumar, V. G., Banerjee, A., and Roy, D. (2025). Prestimulus Periodic and Aperiodic Neural Activity Shapes McGurk Perception. *eNeuro*, 12(10). https://doi.org/10.1523/ENEURO.0431-24.2025

Singh, V. A., Kumar, V. G., Banerjee, A., and Roy, D. (2022). Characterizing Variability of Audiovisual Speech Perception Based on Periodic and Aperiodic Features of Prestimulus Brain Activity. *bioRxiv*. 10.1101/2022.01.20.477172v1

In Prep

**contributing equally*

Singh, V. A., Kumar, V. G., Banerjee, A., and Roy, D. (*in prep*). Prestimulus Large-Scale Connectivity Patterns Predispose Multisensory Perception.

Bhunia, D.*, **Singh, V. A.***, and Roy, D. (*in prep*). A Thalamocortical Neural Field Model Better Elucidate Dynamics of Multisensory Perception. *(Co-leading analysis and drafting)*

Deb, S., Bhunia, D., **Singh, V. A.**, Muralidharan, V., and Roy, D. (*in prep*). Modulation of the Speed of Intrinsic Alpha Frequency (IAF) with tACS Modulates Variability of Illusory Speech Perception. *(Analysis and drafting)*

Singh, V. A.*, P.V*, Jain P*, Singhal T*, Singh P*, *et al.*, Banerjee A., *(in prep)* Multimodal MRI Assessment of Accelerated Cognitive Decline in Rural India: Benchmarking Against the ADNI dataset. *(Co-leading analysis and drafting)*

Research Experience

"Advancing Approaches for Dementia Associated Research (AADAR)" *NBRC*
Dr. Arpan Banerjee (Supervisor) & INCLEN Trust (Collaborator) *April 2025 - Present*
- Co-leading SIEMENS PRISMA 3T–MRI acquisition, quality control (MRIQC), data curation (BIDS), and MRI/ fMRI data pre-processing and SC-FC analysis for a demographically underrepresented older-age cohort in Northern India (To date: N = 359, 40 dementia, 123 MCI, and 196 normal; ages: 55-101),including T1-MPRAGE, T2-SPACE, T2-GRE, FLAIR, DTI, rs-fMRI, and PASL.
- Co-authored a continuation grant proposal (under review) for the AADAR dementia project in collaboration with my PhD supervisor. Drafted the core methodology and scientific rationale *(see Grants section)*.

"Modulation of the Speed of Intrinsic Alpha Frequency (IAF) with tACS Modulates Variability of Illusory Speech Perception" *IIT Jodhpur & NBRC*
Advisors: Dr. Dipanjan Roy, Dr. Vignesh Muralidharan *2024-Present*
- Contributed to McGurk perception experiment design: 64-Channel EEG acquisition planning, and tACS stimulation protocol design; Co-led behavioral data analysis (rm-ANOVA, GLM, Bayesian) and IAF estimation (EEG pre-processing and power spectrum analysis); Result interpretation in collaboration with first authors.

"A Thalamocortical Neural Field Model Better Elucidates the Dynamics of Multisensory Perception" — *IIT Jodhpur & NBRC*

Advisor: Dr. Dipanjan Roy — *2024-Present*

- Contributed to empirical data analysis of 3 datasets (McGurk-EEG, SIFI-EEG, & McGurk-iEEG) and design of the thalamocortical neural-field model, result interpretation, and paper writing in collaboration with first author (Co-author role).

"Comparitive mapping of common mental disorders (CMD) over lifespan" — *NBRC*

Dr. Arpan Banerjee (Supervisor) — *2022-2024*

- Standardized simultaneous EEG-fMRI protocol and preprocessing pipeline for passive moving watching task on SIEMENS 3T PRISMA MRI machine with 64-Channel passive electrode EEG system.

"The Brain Before Perception: Trial-Wise Prestimulus Neural Dynamics Shape Multisensory Perception" — *NBRC*

Advisors: Dr. Arpan Banerjee (Supervisor), Dr. Dipanjan Roy (Co-Supervisor) — *2020-Present*

- My PhD thesis work wherein I have identified EEG spectral, cross-spectral, and functional connectivity patterns at sensor and source space that predict response to multisensory stimulus using mixed-effect regression models, Granger-Geweke Causality, and Bayesian inference.

"Characterizing variability of audiovisual speech perception based on prestimulus oscillatory features of electrophysiological brain signals" — *NBRC*

Advisor: Dr. Dipanjan Roy (Supervisor) — *2019-2020*

Grants & Funding

ICMR Intermediate Grant	**Co-Author / Contributor**, Characterizing early diagnostic markers for progression into dementia from mild cognitive impairments (MCI) and healthy aging in rural monolingual populations: A multi-site study (PI: Dr. Arpan Banerjee)	*Under Review (2026)- Passed 1st Round*

Research Skills & Technical Expertise

Experiment designing	Presentation software and Psychopy.
MRI / fMRI 3T acquisition	T1-MPRAGE, T2-SPACE, T2-GRE, FLAIR, PASL, DTI, rs-fMRI, BIDS curation
EEG acquisition & analysis	Active 64-channel EEG acquisition, preprocessing, artifact rejection, time–frequency analysis, power-spectrum, aperiodic fitting, coherence, PLV, Cross-frequency coupling, source analysis & reconstruction, & Atlas-based parcellations
Simultaneous EEG–fMRI	Acquisition and analysis: synchronization and hardware integration, MRI-related EEG artifact rejection (gradient and pulse), multimodal data integration.
Statistical analysis	Mixed-effects regression models (R, python), rmANOVA (MATLAB, R), Bayesian statistics (R), correlation (R), LASSO regression (R, python), Granger causality (MATLAB), permutation testing (MATLAB, R, python), Drift-diffusion model (python).
Machine learning	GLMs (linear, logistic), Classification (SVM, k-NN), RSA, Dimensionality reduction (PCA, t-SNE), Graph-based Network, cross-validation; Familiarity with: predictive neural decoding and encoding model
Programming	MATLAB, Python, and R.
Toolboxes	BrainVision RecView, EEGLAB, ERPLAB, GEDAI, FieldTrip, MNE, Chronux, Specparam (FOOOF), NBS, FMRIB, fMRIPrep, MRIQC, Nilearn.

Awards, Achievements & Fellowships

2025	**NBRC Travel Grant to attend OHBM 2025, Brisbane**, National Brain Research Centre (NBRC), Manesar	*Rs.75,000*

2024	**EMBO Travel Grant**, Barcelona Summer School for Advanced Modeling of Behavior (BAMB!)	*1,000 euros*
	Journal of Cognitive Neuroscience (JoCN) Travel Award, Cognitive Neuroscience Society (CNS), Toronto, Canada	*3,000 USD*
2023	**OHBM 2023 Travel Award**, Organization for Human Brain Mapping (OHBM)	*~4,000 USD*
2020-Present	**NBRC PhD in Neuroscience Fellowship**, National Brain Research Centre, Manesar	
2018-2020	**NBRC Masters in Neuroscience Fellowship**, National Brain Research Centre, Manesar	
2017	**Certificate of Merit for pursuing The Honors Degree in Zoology**, St. Xavier's College, Mumbai	
2016	**Summer Research Fellow (SRF)**, Indian Academy of Sciences (IAS)	

Presentations

Invited Talks

Singh, V. A., Kumar, V. G., Banerjee, A., and Roy, D. 2026. Prestimulus large-scale directed connectivity patterns predispose multisensory perception. SfN 2026 Nanosymposium, session: *"Human Audiovisual Integration"*, DC, USA *(Upcoming)*

March 2024. *Neuroscience: the past, present and the future*. Invited talk: National Science Day, Birla Balika Vidyapeeth (my alma mater), Pilani, Rajasthan, India.

February 2022. *Role of prestimulus activity in characterizing variability in multisensory perception*. Invited talk: CogSci Seminar Series, hosted by Dr. Vishnu Sreekumar, International Institute of Information Technology (IIIT) Hyderabad, India.

Singh, V. A., Kumar, V. G., Banerjee, A., and Roy, D. 2021. Characterizing Variability of Audiovisual Speech Perception Based on Prestimulus Oscillatory Features of Electrophysiological Brain Signals. Oral presentation: 7th Annual Conference of Cognitive Sciences (ACCS7) 2021, Online

Poster Presentations

** presenting author; + mentored postgraduate*

Singh, V. A., Kumar, V. G., Banerjee, A., and Roy, D. 2025. Prestimulus Whole-Brain Functional Connectivity Underpins McGurk Perception. Organization for Human Brain Mapping (OHBM) 2025, Brisbane, Australia. *Won a Travel Award.*

Singh, V. A., Kumar, V. G., Banerjee, A., and Roy, D. 2024. Predicting Response to McGurk Effect Based on Periodic and Aperiodic Prestimulus EEG Activity. Cognitive Science Society (CogSci) 2024, Rotterdam, Netherlands.

Singh, V. A., Kumar, V. G., Banerjee, A., and Roy, D. 2024. Predicting Response to McGurk Effect Based on Periodic and Aperiodic Prestimulus EEG Activity. Cognitive Neuroscience Society (CNS) 2024, Toronto, Canada. *Won a Travel Award.*

Singh, V. A., Kumar, V. G., Banerjee, A., and Roy, D. 2023. Predicting Response to McGurk Effect Based on Periodic and Aperiodic Prestimulus EEG Activity. Organization for Human Brain Mapping (OHBM 2023), Montreal, Canada. *Won a Travel Award.*

Singh, V. A., Kumar, V. G., Banerjee, A., and Roy, D. 2022. Characterizing Variability of Audiovisual Speech Perception Based on Prestimulus Oscillatory Features of Electrophysiological Brain Signals. 9th Annual Conference of Cognitive Sciences (ACCS9) 2021, New Delhi, India

Singh, V. A., Kumar, V. G., Banerjee, A., and Roy, D. 2020. Characterizing Variability of Audiovisual Speech Perception Based on Prestimulus Oscillatory Features of Electrophysiological Brain Signals. Bernstein Conference 2020, Online.

Deb, S.*+, **Singh, V. A.**, Muralidharan, V., and Roy, D. 2025. Modulation of the Speed of Intrinsic Alpha Frequency (IAF) with tACS Modulates Variability of Illusory Speech Perception. 12th Annual Conference of Cognitive Sciences (ACCS12) 2025, Allahabad, Uttar Pradesh, India.

Bhunia, D.*, **Singh, V. A.**, and Roy, D. 2026. Thalamocortical Neural Field Model for Inter-individual and Inter-Trial Variability in Multisensory Speech Perception. BraiNDy'26, Udaipur, Rajasthan, India.

Teaching Experience

July 2026	**Computational Neuroscience Course**, TA, Neuromatch Academy (NMA) Summer School	*Online*
Fall 2023	**Biostatistics Course**, Graduate TA	*NBRC, India*
Spring 2023	**Cognitive Neuroscience Course**, Graduate TA	*NBRC, India*
July 2022	**Computational Neuroscience Course**, Project TA, NMA Summer School	*Online*

Mentoring

** mentored (under Co-PI supervision); ⁺ Lead mentor*

2024-Present	**Shreya Deb***, MS by Research, Indian Institute of Technology (IIT) Jodhpur
August-September 2023	**Bharti Upparu⁺**, Mentee, WISR (Women in STEM Research) mentorship program, *Guided on building a research career in Cognitive Psychology*
Summer 2023	**Antarjot Kaur***, Summer Intern, Indian Institute of Technology (IIT) Jodhpur, *Won best internship award*
Summer 2023	**Devanshi Singh***, Summer Intern, Indian Institute of Technology (IIT) Jodhpur
Summer 2022	**Palak Sharma***, Summer Intern, Indian Institute of Technology (IIT) Jodhpur, *Won best internship award*

Other Research Experience

"Analysis of the Sex-Ratio and Body Length of *Carcharhinus limbatus* (Blacktip Shark) Over Varied Seasons" — *Mumbai, India*

Research Volunteer, Mumbai Shark Catches Monitoring Programme, Sassoon Dock — *2015-2017*

- Nature Conservation Foundation
- Assisted in data collection, compilation, and analysis of the blacktip shark catches that arrived at the Sassoon dock, Mumbai

"Female Mate Choice Based on Male Age: Are older or younger males better mates in the primitively eusocial wasp *Ropalidia marginata* (paper wasps)?" — *Bangalore, India*

Supervisor: Professor Raghavendra Gadagkar — *May-June 2016*

- Centre for Ecological Sciences (CES), Indian Institute of Science Bangalore (IISc)
- Studied the mate selection behavior in female paper wasps

Outreach & Professional Development

Service & Outreach

January 2026	**Scientific Committee Member**, BraiNDy 2026, Udaipur
December 2025	**Instructor for MRI and EEG hands-on**, Neuroimaging Conference - Clinical and Preclinical Perspectives 2025, National Brain Research Centre (NBRC)
2023-2024	**Fellow**, OHBM Student PostDoc Special Interest Group (OHBM SP-SIG) Fellowship Programme
December 2023	**Volunteer**, Symposium on Brain, Behavior, & Society, National Brain Research Centre (NBRC)
August 2023	**Mentor**, WISR - Women in STEM Research Mentorship Program
July 2023	**TA**, "The Brain Workshop" organized by the Artificial Intelligence and Data Science (AIDE) School of Indian Institute of Technology (IIT), Jodhpur
July 2022	**Project TA**, Computational Neuroscience, Neuromatch Academy (NMA) Online Summer School
2022-2023	**Committee Member**, NBRC Journal Club
2022, 2023	**Scientific Representative of NBRC at MegaScience Expo**, Indian International Science Fest (IISF) 2022 and IISF 2023

Summer Schools

Barcelona Summer School for Advanced Modeling of Behavior (BAMB!), July 2024, Barcelona, Spain

Neuromatch Academy (NMA), Computational Neuroscience, July 2020, Online.

Conference Reviewer

NeuroAI 2026, OpenReview.net

Work Experience

Research Consultant at Moment (now called The Biohackers Co.), April 2022 - December 2022

References

Dr. Arpan Banerjee, Scientist VI & Professor, NBRC, arpan@nbrc.ac.in, +91-124-2845410

Dr. Dipanjan Roy, Professor, IIT Jodhpur, droy@iitj.ac.in, +91-291-2801757